Scaling to Very Very Large Corpora for Natural Language Disambiguation

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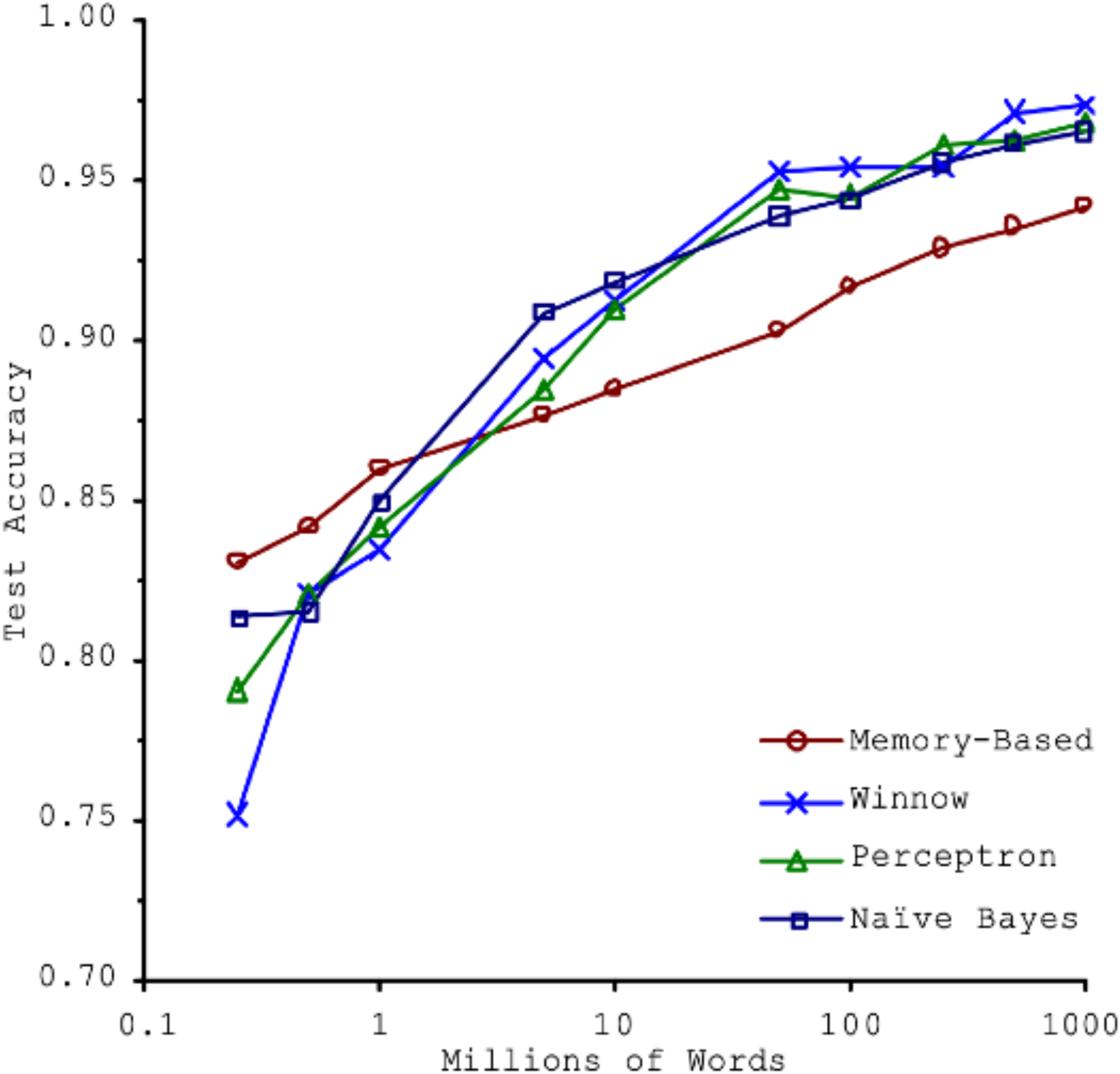
Abstract

The amount of readily available on-line text has reached hundreds of billions of words and continues to grow. Yet for most core natural language tasks, algorithms continue to be optimized, tested and compared after training on corpora consisting of only one million words or less. In this paper, we evaluate the performance of different learning methods on a prototypical natural language disambiguation task, confusion set disambiguation, when

potentially large cost of annotating data for those learning methods that rely on labeled text.

The empirical NLP community has put substantial effort into evaluating performance of a large number of machine learning methods over fixed, and relatively small, data sets. Yet since we now have access to significantly more data, one has to wonder what conclusions that have been drawn on small data sets may carry over when these learning methods are trained using much larger corpora.

In this paper, we present a study of the effects of data size on machine learning for natural language disambiguation. In particular, we study the problem of selection among



Test Accuracy
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.1
1
10
100
1000
Millions of Words
Memory-Based
Winnow
Perceptron
Naïve Bayes

"… We have shown that for a prototypical natural language classification task, the performance of learners can benefit significantly from much larger training sets. We have also shown that both active learning and unsupervised learning can be used to attain at least some of the advantage that comes with additional training data, while minimizing the cost of additional human annotation."

CASE CLOSED

DOES MORE DATA (EXPERIENCE) LEAD TO BETTER PERFORMANCE?

21

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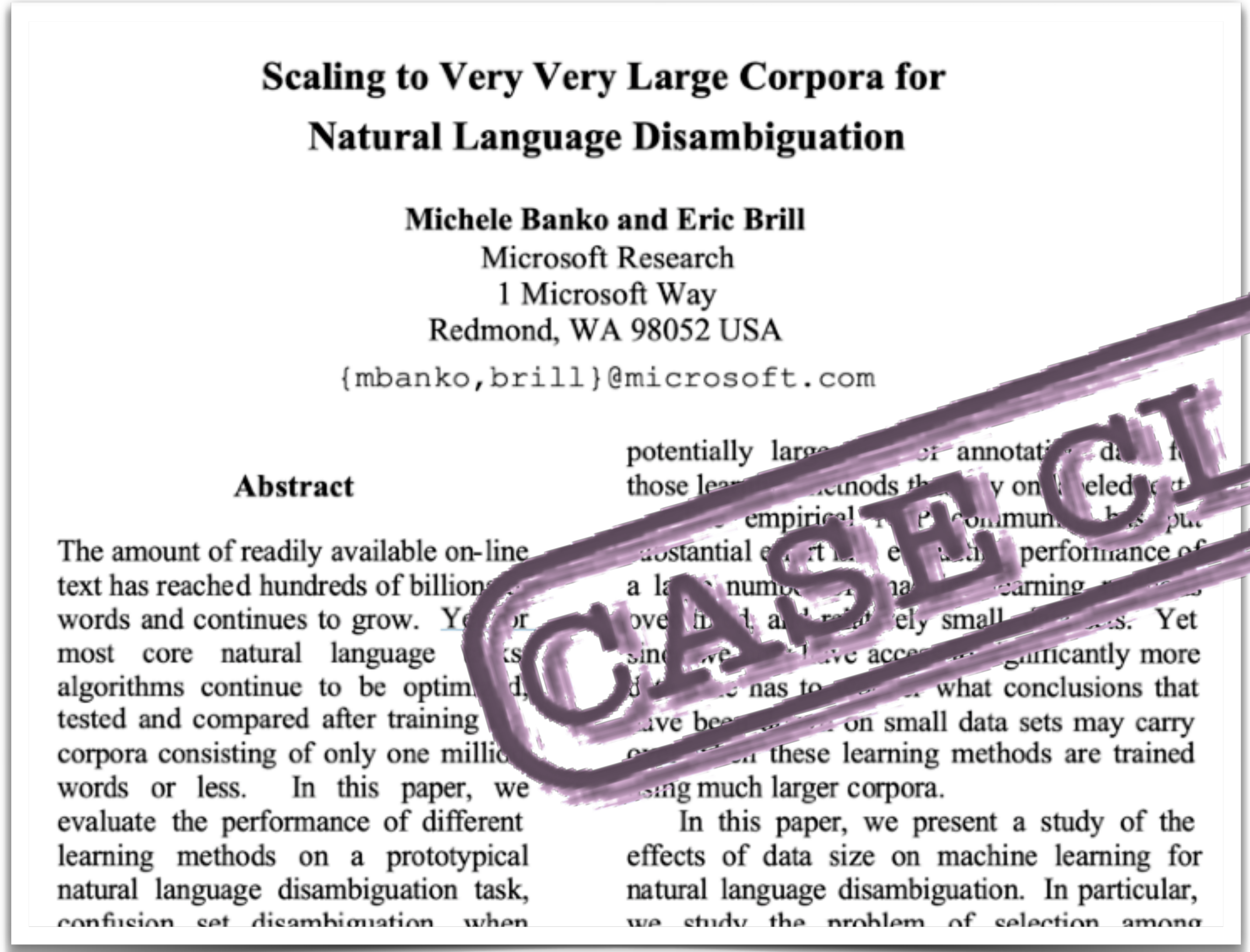

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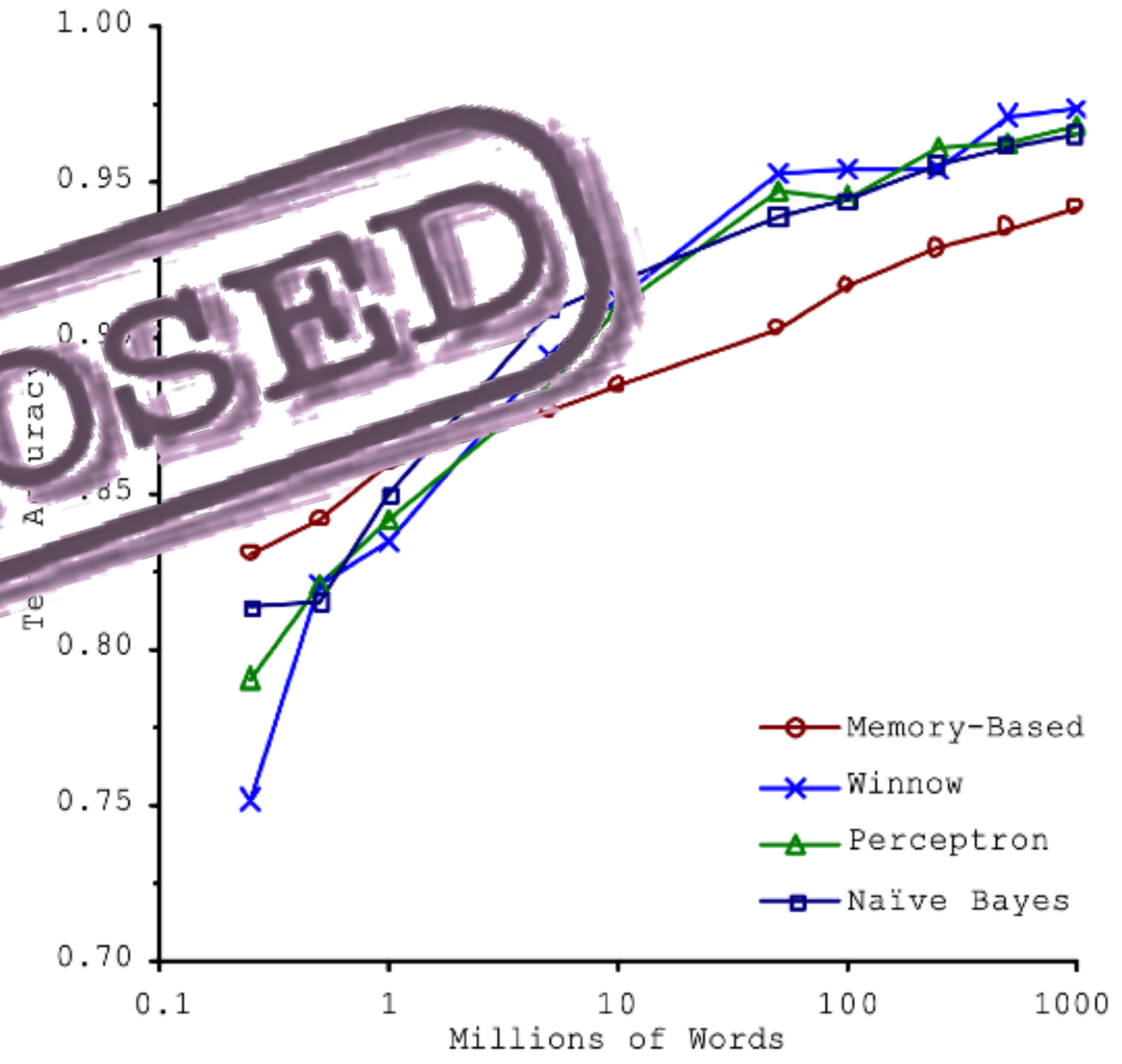

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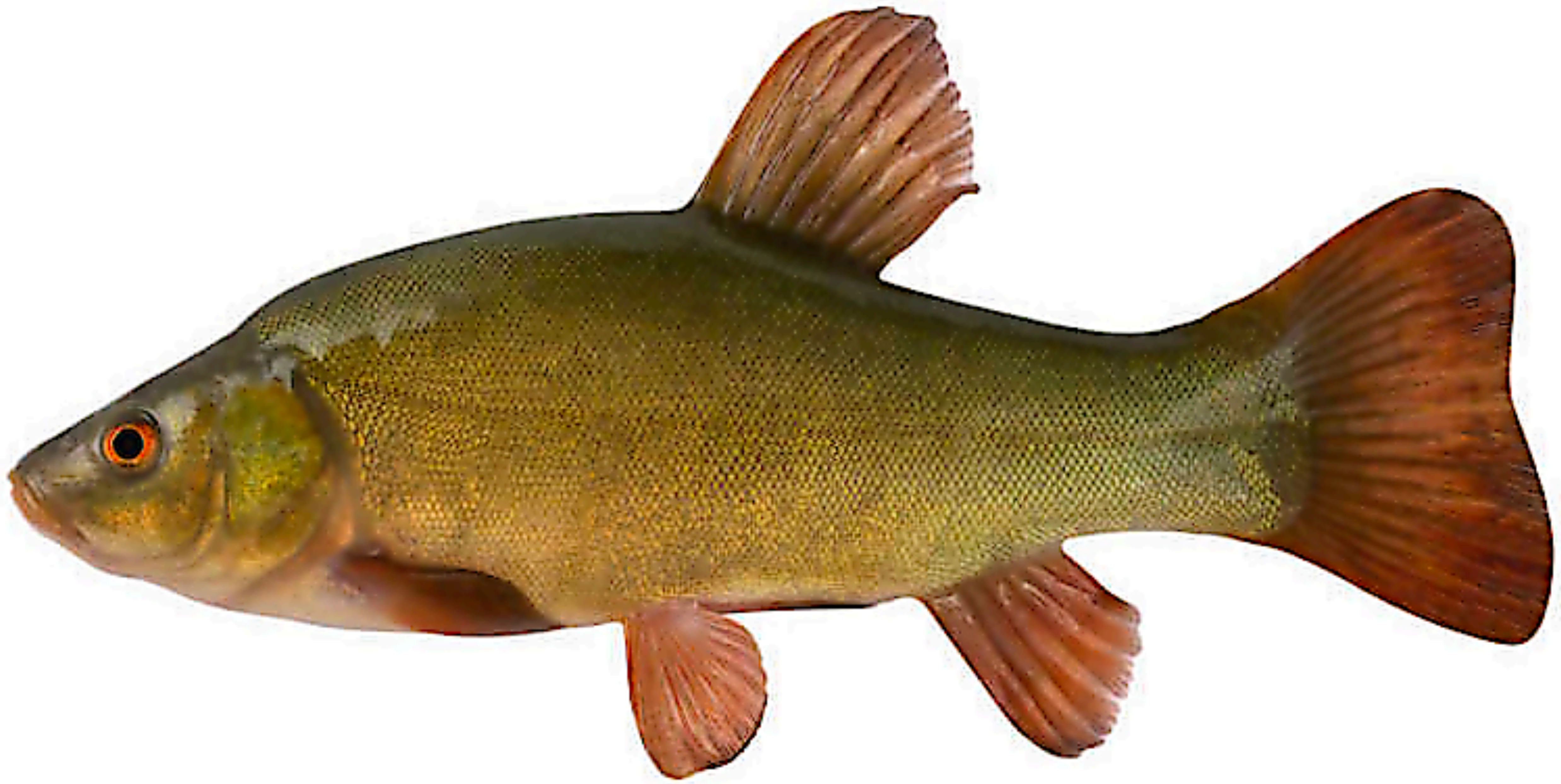
HIVE Lab Health Informatics, Visualization, and Equity
Institute of Health Policy, Management and Evaluation UNIVERSITY OF TORONTO
Dalla Lana School of Public Health

MORE DATA OR BETTER DATA?



Train an AI to identify this tench in pictures!

